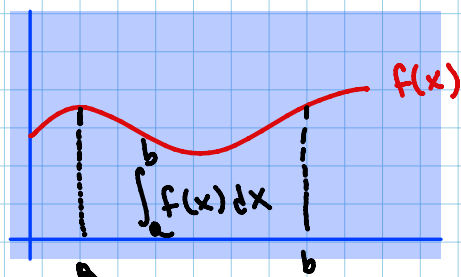


Clase 11 - Integración numérica

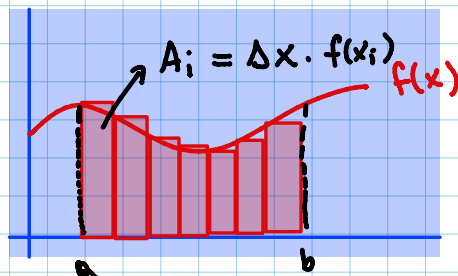
Idea fundamental: Área bajo la curva

$$Ej: f(x) = 3x^2 \rightarrow \int_0^2 f(x) dx = \int_0^2 3x^2 dx = x^3 \Big|_0^2 = 2^3 - 0^3 = 8 //$$



$$A = \int_a^b f(x) dx$$

Riemann
 $\Rightarrow$



$$\therefore A \approx \sum_{i=0}^{n-1} A_i = \sum_{i=0}^{n-1} \Delta x f(x_i)$$

$$\Rightarrow \int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \Delta x f(x_i) \quad ; \quad \Delta x = \frac{b-a}{n} \quad , \quad x_i = a + i \Delta x$$

En nuestro caso:

$$\int_0^2 3x^2 dx = 8 \quad ; \quad \Delta x = \frac{2-0}{n} = \frac{2}{n} \quad ; \quad x_i = 0 + i \cdot \frac{2}{n} \rightarrow x_i = \frac{2i}{n}$$

luego:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \Delta x f(x_i) \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \frac{2}{n} \cdot 3x_i^2 \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \frac{2}{n} \cdot 3 \left(\frac{2i}{n} \right)^2 \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \frac{24i^2}{n^3} \end{aligned}$$

$$= 24 \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{i=0}^{n-1} i^2$$

Sumatoria notable:

$$\sum_{i=0}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\begin{aligned} \text{sea } n \rightarrow n-1: \sum_{i=0}^{n-1} i^2 &= \frac{(n-1)(n-1+1)(2(n-1)+1)}{6} \\ &= \frac{n(n-1)(2n-1)}{6} \end{aligned}$$

$$= 24 \lim_{n \rightarrow \infty} \frac{1}{n^3} \cdot \frac{n(n-1)(2n-1)}{6}$$

$$= 4 \lim_{n \rightarrow \infty} \frac{(n-1)(2n-1)}{n^2}$$

$$= 4 \lim_{n \rightarrow \infty} \frac{2n^2 - n - 2n + 1}{n^2}$$

$$= 4 \lim_{n \rightarrow \infty} \frac{2n^2 - 3n + 1}{n^2}$$

$$= 4 \lim_{n \rightarrow \infty} \left(2 - \frac{2}{n} + \frac{1}{n^2} \right)^0 = 4 \cdot 2 = 8$$

A nivel computacional:

$$\int_a^b f(x) dx \approx \sum_{i=0}^{n-1} \Delta x \cdot f(x_i), \text{ con } n \text{ finito.}$$

¡No podemos sumar infinitas cosas en el computador!

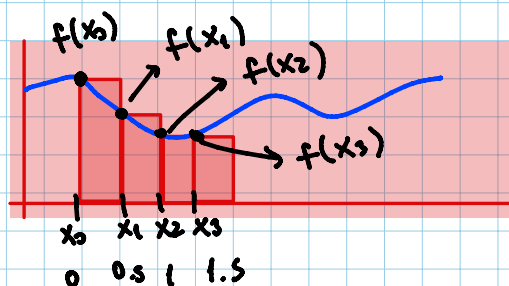
Ejemplo: Sea $n=4$:

$$\int_0^2 3x^2 dx = \sum_{i=0}^{4-1} \Delta x f(x_i); \quad \Delta x = \frac{b-a}{n} = \frac{2-0}{4} = \frac{1}{2} = 0.5$$

$$x_i = a + i \Delta x = 0 + i \cdot 0.5 = 0.5i$$

$$\begin{aligned} \therefore \int_0^2 3x^2 dx &= \sum_{i=0}^3 0.5 f(x_i) = 0.5 \left(f(x_0) + f(x_1) + f(x_2) + f(x_3) \right) \\ &= 0.5 \left(f(0) + f(0.5) + f(1) + f(1.5) \right) \end{aligned}$$

$\underset{0}{} \quad \underset{0.75}{} \quad \underset{3}{} \quad \underset{6.75}{}$



$$= 0.5 \times 10.5 = 5.25 = I_{\text{aprox}}$$

$$\left. \begin{array}{l} I_{\text{real}} = 8 \\ I_{\text{aprox}} = 5.25 \end{array} \right\} \begin{array}{l} \text{Error} \\ \varepsilon_{\%} = 100 \left| \frac{I_{\text{real}} - I_{\text{aprox}}}{I_{\text{real}}} \right| \end{array}$$

$f(x_i)$: altura del rectángulo i -ésimo.

$$\varepsilon_{\%} = 100 \left| \frac{8 - 5.25}{8} \right| = 34.38\%$$

demasiado!!

Conclusión: n debe ser más grande, por ejemplo: 10^5

En la próxima clase estudiaremos el método del Trapecio.

En vez de sumar áreas de Rectángulos... □

Sumamos áreas de trapecios ▭ ✓ Es más preciso.

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